Building a Facial Expression Recognition App Using TensorFlow.js

Let's learn how to build a facial expression recognition app on the TensorFlow.js framework.

TensorFlow is a free and open source software library for data flow and differentiable programming across a range of tasks. It is a symbolic math library, also used for machine learning applications such as deep learning and neural networks. It is used by Google, both in research and in production. It was developed by the Google Brain team for internal use only. On November 9, 2015, it was released under the Apache Licence 2.0 for public use.

TensorFlow can run on multiple CPUs and GPUs (with optional CUDA and SYCL extensions for general purpose computing on GPUs). It is available on 64-bit Linux, MacOS, Windows and mobile computing platforms including Android and iOS. Its flexible architecture allows for the easy deployment of computation across a variety of platforms (CPUs, GPUs and TPUs), and from desktops to clusters of servers, as well as mobile and edge devices.

TensorFlow provides stable APIs for Python and C, without the API backward-compatibility guarantee for C++, Go, Java, JavaScript and Swift.

TensorFlow.js

TensorFlow.js is an open source hardware accelerated JavaScript library for training and deploying machine learning models.

Features

- Develops machine learning (ML) in the Web browser